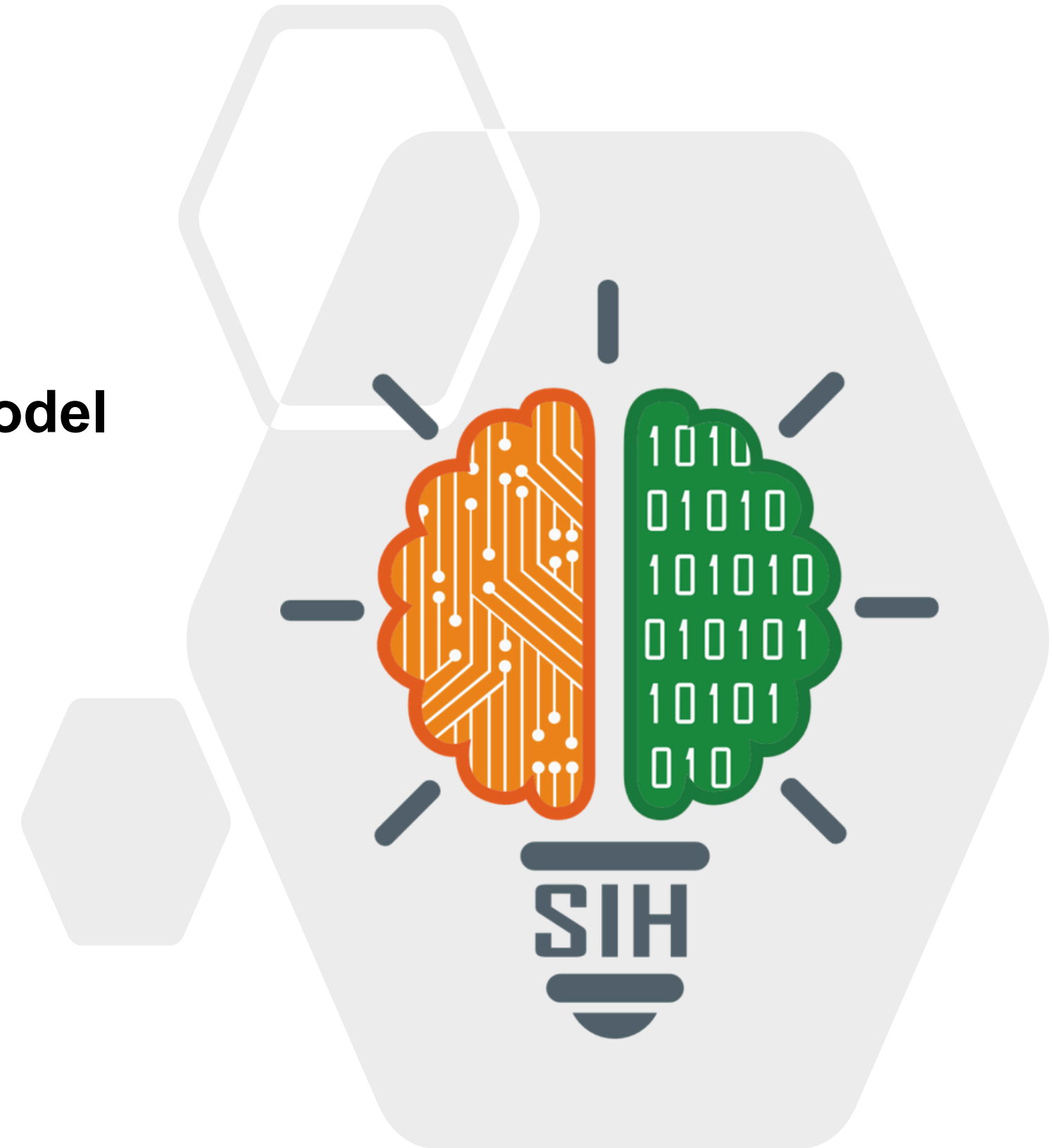


VARUNA

- Problem Statement ID : 26081
- Problem Statement Title : Hybrid AI–NWP Multi-Model

Forecast Blending System

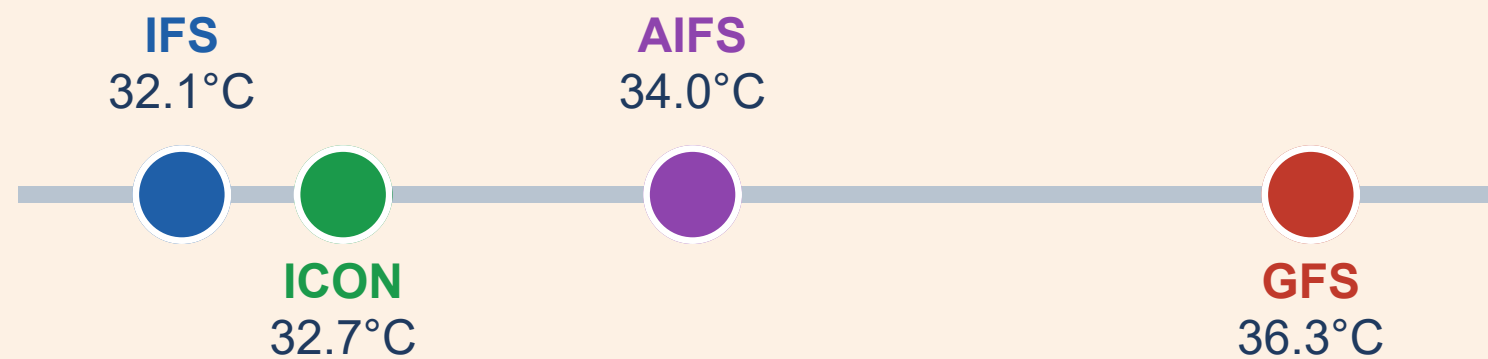
- Theme : Disaster Management
- PS Category : Software
- Team ID : 177105
- Team Name (Registered on portal) :



PROBLEM & SOLUTION

Weather models disagree. Decisions still need one forecast.

THE PROBLEM



MODEL DISAGREEMENT: 4.2°C spread

Which forecast should be trusted?

- 1 Different NWP models give **different forecasts** for the same location and horizon.
- 2 Performance varies with **region, weather regime and lead time**.
- 3 Simple averaging assumes **all models are equally reliable**.
- 4 Operators need **a forecast and the evidence behind it**.

OUR SOLUTION: VARUNA

IFS

AIFS

GFS

ICON

Context + Model Error Prediction

Adaptive Weights

VARUNA Blend

Forecast + Spread + Attribution

WHAT VARUNA PROVIDES

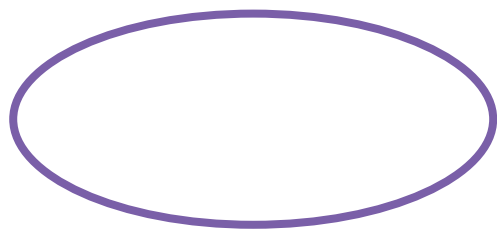
- ✓ Multi-model forecast comparison
- ✓ Contextual adaptive weighting
- ✓ Explainable model contribution
- ✓ Regional forecast intelligence
- ✓ Uncertainty via model spread
- ✓ Extreme-weather surveillance
- ✓ Interactive geo-visualization

TECHNICAL APPROACH

SYSTEM WORKFLOW

TECH STACK

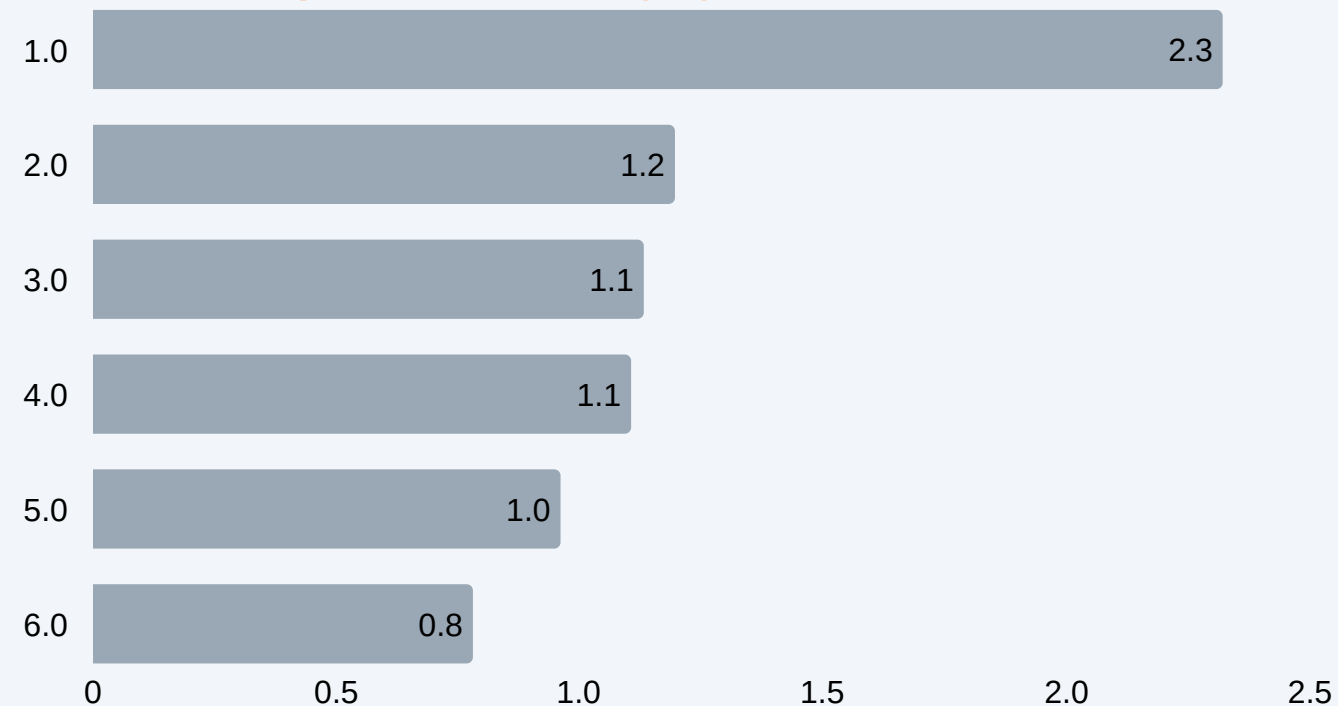




FEASIBILITY AND VIABILITY

Feasibility

Held-out temperature RMSE ($^{\circ}\text{C}$), lower is better



- ✓ Working prototype: FastAPI backend + 8-page dashboard
- ✓ Free, open data: Open-Meteo (CC BY 4.0), ERA5
- ✓ Light model: 4 small XGBoost regressors, no GPU needed
- ✓ 54 backend + 63 data-verification tests in the repo

Challenges and risks

Skill is measured against ERA5 reanalysis, not station observations

Only temperature is trained and validated so far

Test covers one season; 6 of 12 regions have no benchmark

Depends on a live weather API that can go offline

Strategies

Add IMD station ingestion (planned); ERA5 caveat shown on every page

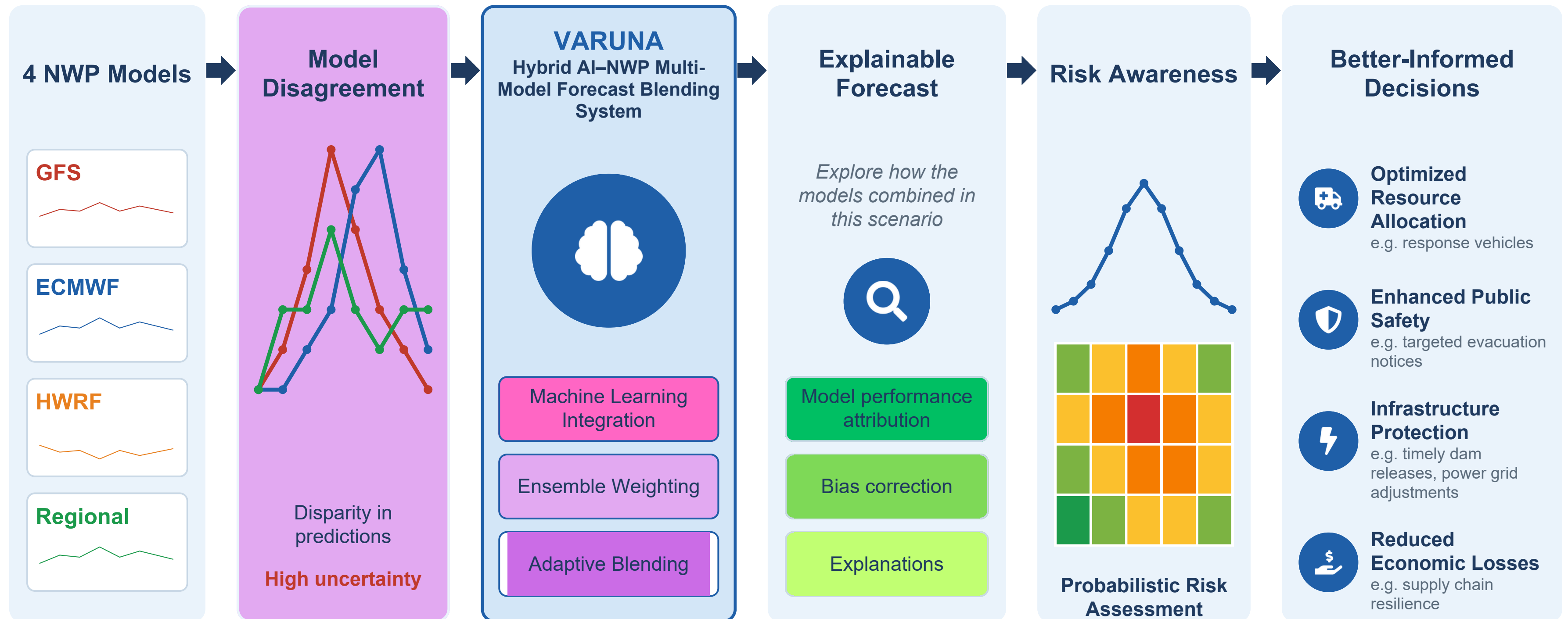
Rain, wind, pressure flagged unvalidated; trained only if they beat the best single model

Extend benchmark windows to all seasons and regions; scope badges show what is proven

30-minute cache plus a clearly labelled REPLAY archive fallback

IMPACT & BENEFITS

Turning *model disagreement* into actionable forecast intelligence



RESEARCH AND REFERENCES

